

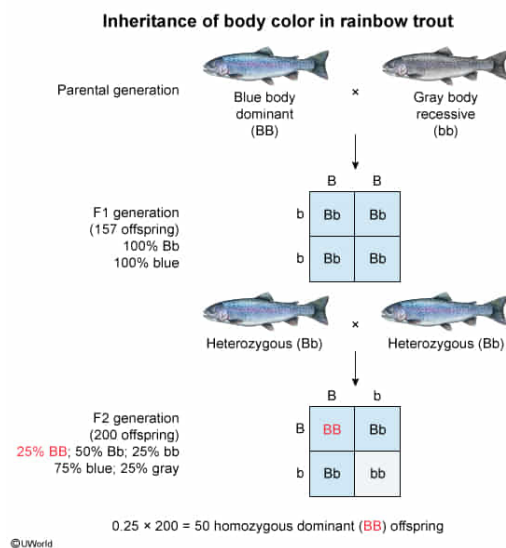
In rainbow trout, body color is inherited in a Mendelian pattern, and blue body color is dominant to gray body color. A blue trout and a gray trout were crossed and produced 157 F1 offspring that were all blue. If two of these F1 offspring were then crossed and produced 200 offspring in the F2 generation, approximately how many F2 trout are homozygous dominant for this trait?

- ☐ A. 0 [2%]
✓ ☒ B. 50 [79%]
☐ C. 100 [11%]
☐ D. 150 [6%]

Correct

78% answered correctly

Explanation:



In a **monohybrid cross**, individuals who are identically **heterozygous** at a specific genetic locus are mated. In this question, the first cross (mating) is performed between a parental pair of rainbow trout, one of which is blue and one of which is gray. The question states that body color is inherited in a Mendelian pattern and that blue body color (B) is **dominant** to gray body color (b).

The results of the first cross are all blue trout. Given this, it is likely that one parent was homozygous dominant for the body color trait (BB) and the other parent was homozygous recessive (bb), creating a completely heterozygous F1 generation (Bb). The question goes on to state that two of the F1 generation were crossed and produced 200 offspring in the F2 generation, and then asks for the approximate number of F2 trout that are homozygous dominant (BB) for the body color trait.

To determine this, the percentage of individuals with each genotype must be calculated. A **Punnett square** is a useful way to visualize this distribution. Crossing two heterozygous (Bb) F1 trout produces an F2 generation with 25% homozygous dominant (BB), 50% heterozygous (Bb), and 25% homozygous recessive (bb) trout. If 200 offspring are produced in the F2 generation, the approximate number of trout that are homozygous dominant for this trait (BB) can be calculated as follows:

Homozygous dominant(BB) F2 trout = $0.25 \times \text{total offspring} = 0.25 \times 200 \text{ trout} = 50 \text{ trout}$

Therefore, approximately **50 F2 trout** are expected to be homozygous dominant (BB) for the body color trait. Because blue body color is dominant to gray, these 50 trout will have **blue bodies**.

(Choices A, C, and D) These options represent *incorrect* calculations of the number of homozygous dominant trout in the F2 generation.

Educational objective:

In a monohybrid cross, members of the first (parental) generation are homozygous for different alleles at a specific locus. The first cross produces an F1 generation of heterozygous offspring. A subsequent cross produces an F2 generation with a mix of genotypes and phenotypes.

Time spent:0 sec

Subject:
Biology

QID:403739

System:
Genetics and Evolution